

```

#include <iostream>
using namespace std;

const int SPECIAL = 13;
int main() {
    int totalOccupied = 0, totalRooms = 0, nFloors, nRooms, occupied;
    double percent;

    cout << "How many floors in the hotel?\n";
    cin >> nFloors;

    /* Set with 0 to avoid garbage value after the loop */
    nRooms = 0;
    occupied = 0;

    for (int i = 1; i <= nFloors; i++) {
        int roomsPerFloor=0, occupiedPerFloor=0;
        cout << "How many rooms on floor " << (i) << "?\n";
        cin >> roomsPerFloor;
        nRooms += roomsPerFloor;

        cout << "How many are occupied?\n";
        cin >> occupiedPerFloor;
        occupied += occupiedPerFloor;
    }
    percent = static_cast<double>(occupied) / static_cast<double>(nRooms) * 100;

    cout << endl;
    cout << "Total rooms: " << nRooms << endl;
    cout << "Occupied rooms: " << occupied << endl;
    cout << "Unoccupied rooms: " << (nRooms - occupied) << endl;
    cout << "% rooms occupied: " << percent << endl;
    cout << endl;
    return 0;
}

```

```
#include <iostream>
#include <fstream>
#include <string>
using namespace std;

int main() {
    ofstream outFile;
    int nStudents, age;
    string name;

    cout << "Number of students: ";
    cin >> nStudents;
    cin.ignore();
    outFile.open("students.txt");

    for (int i = 0; i < nStudents; i++) {
        cout << "Student " << i+1 << " name: ";
        getline(cin, name);

        cout << "Name: " << name << endl;

        cout << "Student " << i+1 << " age: ";
        cin >> age;
        cin.ignore();
        outFile << name << " " << age << endl;
    }
    return 0;
}
```

```
hotel.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  const int SPECIAL = 13;
5  int main() {
6      int totalOccupied = 0, totalRooms = 0, nFloors, nRooms, occupied;
7      double percent;
8
9      cout << "How many floors in the hotel?\n";
10     cin >> nFloors;
11
12     /* Set with 0 to avoid garbage value after the loop */
13     nRooms = 0;
14     occupied = 0;
15
16     for (int i = 1; i <= nFloors; i++) {
17         int roomsPerFloor=0, occupiedPerFloor=0;
18         cout << "How many rooms on floor " << (i) << "?\n";
19         cin >> roomsPerFloor;
20         nRooms += roomsPerFloor;
21
22         cout << "How many are occupied?\n";
23         cin >> occupiedPerFloor;
24         occupied += occupiedPerFloor;
25     }
26     percent = static_cast<double>(occupied) / static_cast<double>(nRooms) * 100;
27
28     cout << endl;
29     cout << "Total rooms: " << nRooms << endl;
30     cout << "Occupied rooms: " << occupied << endl;
31     cout << "Unoccupied rooms: " << (nRooms - occupied) << endl;
32     cout << "% rooms occupied: " << percent << endl;
33     cout << endl;
34     return 0;
35 }
```

```
studentdb.cpp > main()
1  #include <iostream>
2  #include <fstream>
3  #include <string>
4  using namespace std;
5
6  int main() {
7      ofstream oFile;
8      int nStudents, age;
9      string name;
10
11      cout << "Number of students: ";
12      cin >> nStudents;
13      cin.ignore();
14      oFile.open("students.txt");
15
16      for (int i = 0; i < nStudents; i++) {
17          cout << "Student " << i+1 << " name: ";
18          getline(cin, name);
19
20          cout << "Name: " << name << endl;
21
22          cout << "Student " << i+1 << " age: ";
23          cin >> age;
24          cin.ignore();
25          oFile << name << " " << age << endl;
26      }
27      return 0;
28 }
```

```
How many floors in the hotel?
5
How many rooms on floor 1?
12
How many are occupied?
7
How many rooms on floor 2?
81
How many are occupied?
50
How many rooms on floor 3?
69
How many are occupied?
69
How many rooms on floor 4?
4
How many are occupied?
2
How many rooms on floor 5?
58
How many are occupied?
10

Total rooms: 224
Occupied rooms: 138
Unoccupied rooms: 86
% rooms occupied: 61.6071
```

```
Number of students: 5
Student 1 name: Ian Fettes
Name: Ian Fettes
Student 1 age: 22
Student 2 name: Bob Gilber
Name: Bob Gilber
Student 2 age: 54
Student 3 name: Ace Ventura
Name: Ace Ventura
Student 3 age: 69
Student 4 name: Adam Sandler
Name: Adam Sandler
Student 4 age: 47
Student 5 name: Nicholas Cage
Name: Nicholas Cage
Student 5 age: 80
```

≡ students.txt

```
1   Ian Fettes 22
2   Bob Gilber 54
3   Ace Ventura 69
4   Adam Sandler 47
5   Nicholas Cage 80
6   |
```